HOLA: Towards Holistic Human–AI Collaborative Framework for Interactive Aortic CTA Segmentation and Quality Assessment

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Abstract. Accurate segmentation of the aorta from computed tomography angiography (CTA) is essential for diagnosis, treatment planning, and quantitative assessment of aortic disease. Manual segmentation is time-consuming and subject to inter-observer variability, while fully automated models can produce errors that are difficult to correct in complex anatomy. We propose HOLA, a three-dimensional (3D) human-AI collaborative framework combining automatic segmentation, interactive volumetric correction, and quality-aware feedback. HOLA integrates a 3D DynUNet with the DeepEdit interaction strategy, using positive and negative click maps to iteratively refine segmentations across the full CTA volume, trained via a progressive multi-dataset strategy across six CTA datasets to improve generalisation across heterogeneous anatomy and pathology. On 72 held-out test cases, mean Dice improved from 0.804 to 0.868 after ten clicks (Wilcoxon signed-rank, $p < 0.001$). A quality prediction network estimated segmentation quality without ground truth ($r = 0.953$, MAE $= 0.039$), enabling a quality-guided stopping criterion achieving 95.3% precision while reducing interaction from 10.00 to 1.03 clicks per case. These results demonstrate the potential of 3D interactive refinement with reference-free quality assessment for efficient human-AI collaborative aortic segmentation.

Keywords: Aortic Segmentation · Computed Tomography Angiography · Interactive Segmentation · Human–AI Collaboration · Deep Learning · Quality Assessment · Aortic Disease

1 Introduction

Aortic diseases, including aneurysm and dissection, carry substantial morbidity and mortality, making accurate assessment of aortic anatomy important for diagnosis and treatment planning [15]. Computed tomography angiography (CTA) provides detailed three-dimensional (3D) visualisation of the aorta, but manual segmentation remains time-consuming and challenging in anatomically complex cases [9].

Deep learning has improved automated segmentation through CNN-based models such as U-Net [13] and nnU-Net [5], as well as transformer-based approaches including UNETR [3] and CIS-UNet [4]. However, fully automated models produce fixed predictions that can be difficult to correct. Foundation models such as SAM [7], Medical SAM2 [17], and MedSAM2 [8] enable prompt-guided interaction, but provide limited aorta-specific optimisation and no integrated mechanism for estimating segmentation quality. Interactive methods such as DeepGrow [14] and DeepEdit [1] allow iterative refinement, yet users still lack an objective indication of when segmentation is sufficiently accurate to stop.

To address these limitations, HOLA provides an end-to-end human-AI workflow combining initial 3D segmentation, positive and negative click-based correction, and automated quality estimation. A 3D DynUNet [5] incorporates both click maps and CTA information to enable volumetric refinement across the full scan, while progressive multi-dataset training exposes the model to diverse aortic pathologies. A quality prediction module further estimates segmentation quality without ground truth during inference, and the resulting score is integrated into a Streamlit interface to provide real-time feedback and stopping guidance.

2 Method

We propose the HOLA framework (Fig. 1), which integrates CTA preprocessing, 3D interactive segmentation, automated quality prediction, and user-guided refinement into a unified workflow. Rather than producing a fixed segmentation or relying solely on local correction, HOLA progressively refines the full volume while estimating segmentation quality to support informed stopping decisions.

2.1 3D Interactive Segmentation

Each CTA volume was standardised before segmentation. HOLA uses a DeepEdit-enabled 3D DynUNet [5, 1] that receives the CTA volume with Gaussian-smoothed positive and negative click maps. At interaction step t,

$$S_t = f_\theta(I, C_t^+, C_t^-), \qquad t = 0, \ldots, T, \tag{1}$$

where I is the CTA volume, C_t^+ and C_t^- are the accumulated click maps, and f_θ is the segmentation network. At $t = 0$, no corrective clicks are provided, producing

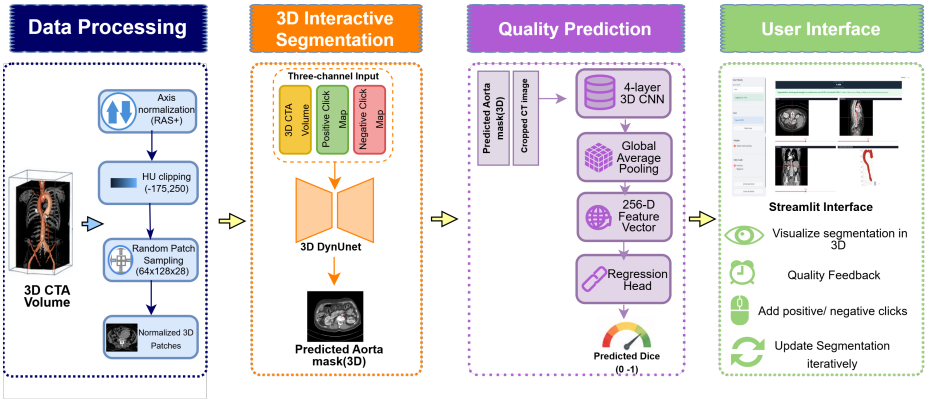


Fig. 1. Overview of the HOLA human–AI collaborative framework for 3D interactive aortic segmentation and quality assessment.

the initial automatic segmentation. The model was trained using combined Dice and binary cross-entropy (BCE) losses:

$$\mathcal{L}_{seg} = \mathcal{L}_{Dice} + \mathcal{L}_{BCE}. \tag{2}$$

During training, DeepEdit simulates user corrections by identifying false-negative and false-positive regions and generating positive and negative clicks, respectively. Multiple error regions are combined into a distance-weighted probability map that prioritises region centres, from which a click is sampled stochastically. This enables the same network to perform both initial segmentation and iterative volumetric refinement.

2.2 Reference-Free Quality Prediction

To provide real-time quality feedback without ground truth, HOLA incorporates a 3D quality prediction network. For each predicted segmentation, a local region containing the segmented foreground and its corresponding CTA context was extracted from the image and binary prediction. The cropped CTA volume I_Ω and predicted segmentation $S_{t,\Omega}$ were provided to the quality prediction network:

$$q_t = c_\phi(I_\Omega, S_{t,\Omega}) \in [0, 1], \tag{3}$$

where q_t denotes the estimated segmentation quality at interaction step t, Ω represents the local region of interest, and c_ϕ denotes the trained quality prediction network. The quality prediction network was trained using mean squared error against true DSC, with interaction samples generated from the model-development cohorts. The stopping threshold τ was calibrated using validation data only. During inference, the predicted quality q_t is compared with τ, with refinement continuing while $q_t < \tau$ and stopping recommended when $q_t \geq \tau$.

The held-out test set was excluded from quality-network training and threshold selection and used only for final evaluation.

2.3 Progressive Multi-Dataset Training

To improve robustness across heterogeneous anatomy, acquisition protocols, and aortic pathologies, the model was trained across six CTA datasets using a fixed validation set ($n = 71$) and held-out test set ($n = 72$) throughout all experiments. The training pool was progressively expanded by introducing the datasets in the following order: Base → Dissection [10] → SEGA [11] → CIS-UNet [4] → AortaSeg-60 [2] → ImageTBAD [16]. The corresponding training sample sizes increased from 27 patients at baseline to 49, 82, 116, 158, and finally 214 patients, with the final configuration comprising 357 patients across the training, validation, and test partitions. This allows the effect of increasing dataset diversity on segmentation performance and generalisation to be systematically evaluated.

The segmentation network was trained using the Adam optimiser with an initial learning rate of 1×10^{-3} and a cosine annealing learning-rate schedule. Early stopping was applied with a patience of 15 epochs. Model checkpoints were selected according to the highest validation DSC obtained after three simulated corrective clicks. All experiments were conducted on the QMUL Apocrita high-performance computing cluster [6].

3 Experiments and Results

This section evaluates HOLA through five complementary experiments: overall and cohort-specific segmentation performance, robustness to imperfect user interactions, reference-free quality prediction, multi-dataset training ablations, and comparison with foundation-model baselines. Quantitative evaluation was performed on 72 held-out cases drawn from six CTA cohorts.

3.1 Overall Segmentation Performance

Evaluated on a held-out test set of 72 cases across six CTA cohorts, HOLA achieved an initial automatic DSC of 0.804 (95% CI: 0.7352–0.8692), improving progressively with user guidance to reach 0.851 at one click, 0.861 at five, and 0.868 after ten interactions (95% CI: 0.8099–0.9188). The 0.0633 overall improvement was statistically significant via a Wilcoxon signed-rank test ($p < 0.001$). Final performance reached IoU (Intersection over Union) of 0.820 and HD95 (95th percentile Hausdorff Distance) of 16.27 mm. The sharpest gain occurred at the first interaction (+0.047), after which incremental benefits diminished as major errors were corrected. These results indicate that the greatest benefit of interaction is obtained from the initial corrections, followed by diminishing improvements as segmentation errors are progressively resolved.

Table 1. Per-cohort segmentation performance of the final HOLA model on the held-out test set ($N = 72$). $\text{Dice}_{\setminus C}$ is the overall test-set Dice with cohort C excluded, and $\Delta_{\text{LOO}} = \text{Dice}_{\setminus C} - \text{Dice}_{\text{all}}$; a larger Δ_{LOO} indicates a cohort disproportionately dragging down the overall average.

Cohort	Pathology	N	$\text{Dice}_{@0} \uparrow$	$\text{Dice}_{@10} \uparrow$	$\Delta\text{Dice} \uparrow$	$\text{Dice}_{\setminus C} \uparrow$	$\Delta\text{LOO} \uparrow$
Base	Aortic Stenosis	5	0.976	0.977	+0.001	0.860	+0.0082
Dissection [10]	Aortic dissections	8	**0.980**	**0.983**	+0.003	0.853	**+0.0144**
SEGA [11]	Aortic dissections	14	0.775	0.917	**+0.142**	0.856	+0.0118
CIS-UNet [4]	Type B Aortic Dissections	12	0.845	0.920	+0.075	0.857	+0.0105
AortaSeg-60 [2]	Aortic dissections	7	0.854	0.871	+0.017	0.867	+0.0004
TBAD [16]	Type-B Aortic Dissections	26	0.702	0.760	+0.058	**0.929**	-0.0612
All cohorts	Combined Test Set	72	0.804	0.868	+0.064	–	–

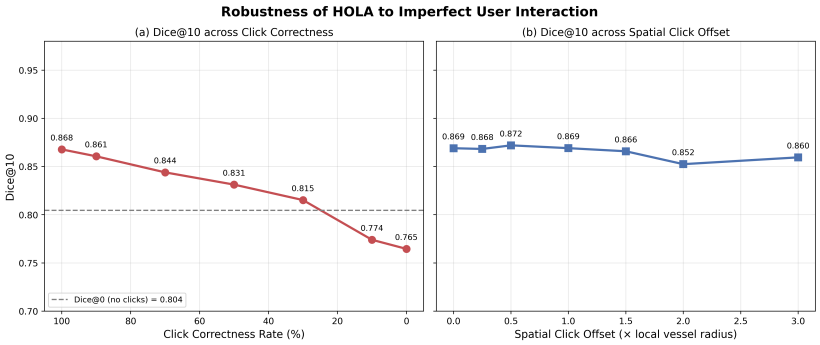


Fig. 2. Robustness of HOLA to spatially imprecise and semantically incorrect user interactions.

3.2 Per-cohort Performance and Generalisation

Table 1 presents the performance of the final HOLA model across the six independent CTA cohorts.Table 1 shows performance across six CTA cohorts. Interactive gains depend on initial quality: SEGA and CIS-UNet improved significantly (+0.142 and +0.075), while Base and Dissection cohorts started high (0.976 and 0.980), leaving little room for improvement. The TBAD cohort offers the key insight: starting lowest at 0.702, interaction improved it to 0.760 (+0.058). This proves interactive correction helps but cannot fully overcome extreme pathological complexity. Leave-one-cohort-out analysis (ΔLOO of -0.0612) confirms TBAD serves as a rigorous test domain rather than a negative influence. Ultimately, TBAD highlights the current boundary of interactive refinement and motivates future pathology-aware modelling.

3.3 Robustness to Imperfect User Interaction

The robustness of HOLA to imperfect interaction was evaluated in terms of spatial click precision and semantic correctness. As shown in Figure 2, HOLA

remained robust to negative-click displacement of up to 3.0× the local vessel radius, with Dice@10 ranging from 0.852 to 0.872 compared with a 0.868 baseline. In contrast, semantic errors had a stronger impact: at 50% correct decisions, Dice@10 decreased to 0.831, while completely incorrect feedback reduced performance below the initial automatic baseline (0.765 vs. 0.804). These results suggest that HOLA is more tolerant of imprecise click placement than incorrect user decisions.

A pilot evaluation using real user click trajectories across 33 test cases further showed a negligible overall change in Dice (mean ΔDice = +0.001). Among 16 multi-click cases, 4 improved, 4 worsened, and 8 showed no change, highlighting the variability of real user interaction compared with the more consistent improvements observed with simulated clicks.

3.4 Quality Prediction Network Evaluation

Evaluating the prediction network against 432 held-out samples revealed strong alignment with true Dice values $r = 0.953$, and MAE = 0.039. At a calibrated stopping threshold of $\tau = 0.60$, the criterion achieved 95.3% precision, 100% sensitivity, 72.3% specificity, and 95.8% accuracy. Retrospective replay across 72 test patients showed that 61 cases (84.7%) stopped at zero clicks, 4 (5.6%) after one click, and only 7 (9.7%) required the full ten-click budget, decreasing the mean interaction to 1.03 clicks. While incurring a minor mean Dice reduction (0.8466 vs. 0.8690, difference of −0.0224; Wilcoxon signed-rank $W = 150$, $p < 0.001$), the 95.3% precision guarantees clinical safety. The quality-guided stopping criterion thus achieves nearly 90% fewer clicks at a modest, statistically confirmed cost to accuracy.

3.5 Ablation Studies

Two complementary ablation studies were conducted to isolate the effects of dataset ordering and training-set size while keeping validation and test distributions fixed.

Impact of Dataset Composition and Integration Order The progressive cohort accumulation strategy was compared with an alternative dataset order: Base → TBAD → CIS-UNet → AortaSeg-60 → Dissection → SEGA.

As shown in Figure 3, the incremental trajectory exhibited sharp performance fluctuations, such as an initial drop upon introducing the Dissection cohort before recovering, whereas the randomised trajectory maintained a smoother, consistently higher intermediate performance. However, as detailed in Table 2, both strategies ultimately converged to comparable final DSC at ten clicks (0.868 vs. 0.871). This demonstrates that dataset integration order influences intermediate optimisation trajectories but has a limited effect on final interactive segmentation performance once the complete multi-cohort training set is incorporated.

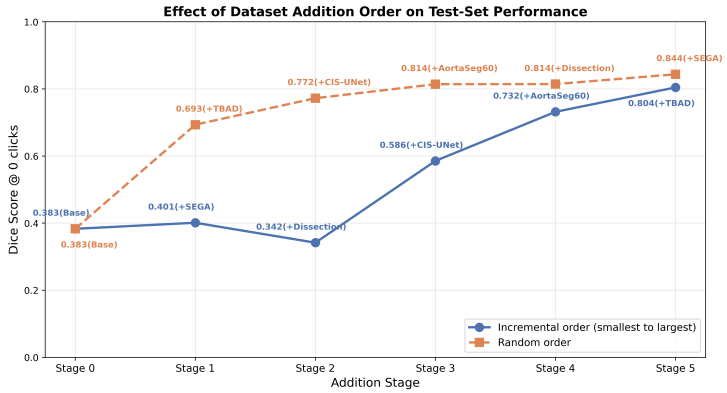


Fig. 3. Multi-cohort training Trajectories under progressive and alternative dataset integration orders.

Table 2. Effect of training-set size and dataset integration order on held-out test performance. IoU and HD95 are reported at 10 clicks.

Model	Dice@0↑	Dice@5↑	Dice@10↑	ΔDice↑	IoU↑	HD95↓
Subset-20	0.629	0.689	0.715	+0.086	0.631	–
Subset-50	0.544	0.678	0.712	**+0.168**	0.620	–
Subset-100	0.789	0.836	0.849	+0.060	0.795	–
Subset-150	0.754	0.813	0.818	+0.064	0.760	–
Subset-200	**0.816**	**0.870**	**0.876**	+0.060	**0.827**	14.92
Incremental order	0.804	0.861	0.868	+0.064	0.820	16.27
Random order	0.844	0.868	0.871	+0.027	0.826	**12.04**

Impact of Training-set Size As shown in Table 2, final performance increased from 0.715 with 20 patients to 0.876 with 200 patients. Small performance drops at 50 and 150 patients show that having more patients does not automatically guarantee better results. Interestingly, weaker starting models like the 50-patient subset gained the most from user corrections (ΔDice = +0.168), proving that interactive clicks can easily make up for a weaker initial segmentation.

As summarised in Table 2, the 200-patient multi-source subset achieved the strongest overall performance (Dice@10 of 0.876, IoU of 0.827). Collectively, these findings demonstrate that comprehensive data diversity and sufficient total volume are far more critical for robust vascular segmentation than the precise historical sequence in which datasets are introduced.

3.6 Comparisons with Foundation Baseline Models

Table 3 compares HOLA with foundation models under varying interaction protocols. AHOLA uses volumetric corrective clicks, whereas MedSAM2-2D uses

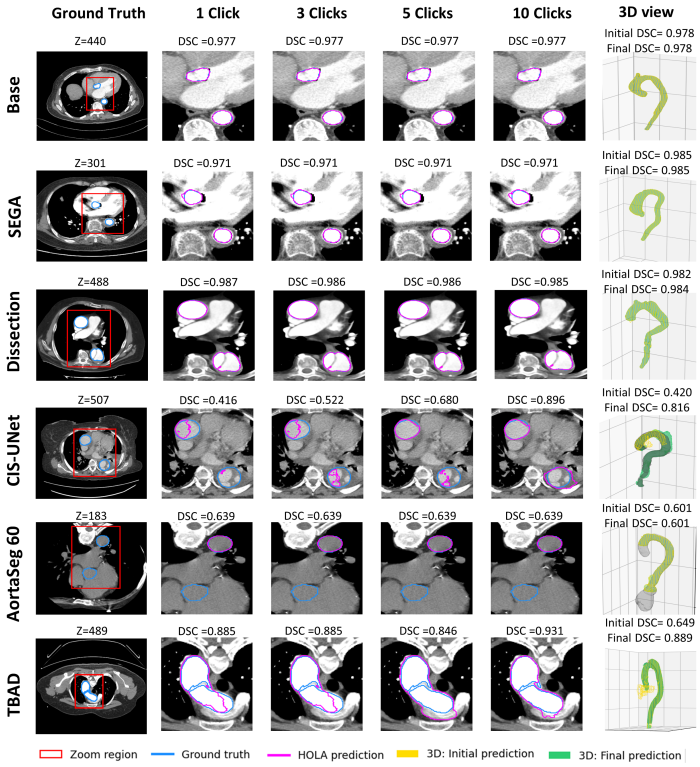


Fig. 4. Representative examples of interactive segmentation refinement across the six CTA cohorts.

slice-wise point prompts. In the matched comparison, HOLA achieved higher Dice than MedSAM2-2D at both zero clicks (0.816 vs. 0.765) and ten clicks (0.880 vs. 0.867). At zero clicks, the median paired Dice difference was +0.057 (95% CI: 0.043–0.090), with a significant Wilcoxon signed-rank test ($W = 613$, $p < 0.001$). After ten clicks, the median paired difference remained +0.042 (95% CI: 0.030–0.048), also significant by Wilcoxon signed-rank testing ($W = 597$, $p < 0.001$). The smaller mean difference at ten clicks (+0.013) indicates that interactive refinement reduced the overall performance gap, while HOLA retained higher performance across the matched cases.

3.7 Cross-Cohort Qualitative Evaluation of Interactive Refinement

Figure 4 illustrates representative cohort-specific refinement behaviours across the interaction sequence. In several cases, including the Baseline, SEGA, and Dissection examples, the segmentation remained stable after the initial prediction because the prediction network had already identified the result as sufficient

Table 3. Comparison of HOLA with foundation-model baselines on the held-out test set. Dice, IoU, and HD95 are reported at each method's final result.

Method	Prompt / interaction	Dice↑	IoU↑	HD95 (mm)↓
MedSAM [7]	Bounding box	0.421	0.290	130.48
MedSAM2 [8]	2D clicks, 10 clicks	0.867	0.789	**15.88**
MedSAM2-3D [12]	2.5D clicks, 10 clicks	0.636	0.497	71.32
HOLA (Ours)	3D clicks, 10 clicks	**0.868**	**0.820**	16.27

quality at the corresponding iteration. The absence of further changes after this point therefore reflects successful quality-guided stopping rather than a failure to improve. In contrast, the CIS-UNet example required further refinement, with 2D Dice increasing from 0.416 to 0.896 and volumetric Dice from 0.420 to 0.816, demonstrating the benefit of corrective interaction when substantial initial errors were present. AortaSeg-60 showed limited improvement, while TBAD demonstrated substantial volumetric improvement despite its challenging pathology. Overall, the examples demonstrate that HOLA can distinguish between cases requiring further correction and cases where the current segmentation is already sufficiently accurate, reducing unnecessary user interaction.

4 Conclusion

This paper presents HOLA, a 3D human-AI collaborative framework for interactive aortic segmentation from CTA, combining volumetric corrective interaction with reference-free quality-guided feedback. Results demonstrated that interaction effectively refines challenging segmentations, while quality-aware stopping substantially cuts down the required workload while maintaining high accuracy. Furthermore, evaluation across diverse datasets highlights that while multi-dataset training improves generalisation, severe pathology like TBAD remains challenging, underscoring the value of integrating proactive refinement with quality-aware estimation to streamline clinical workflows.

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